

Task 1:

```
public class Task1{
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello, group 2");
        sb.append("! Welcome ");
        sb.append("to COMP ");
        sb.append(251);
        sb.append(" course.");
        System.out.println(sb);
    }
}
```

Output: Hello, group 2! Welcome to COMP 251 course.

Task 2:

```
public class Task2{
    public static void main(String[] args) {
        String s1 = "135";
        String s2 = "215";

        System.out.println("Sum of " + s1 + " and " + s2 + " is: " +
(Integer.parseInt(s1) + Integer.parseInt(s2)));
    }
}
```

Output: Sum of 135 and 215 is: 350

Task 3:

```
public class Task3{
    public static void main(String[] args) {
        String s1 = "Hello";
        String s2 = s1;
        String s3 = new String("Hello");

        System.out.println(s1 == s2);           //true because the compiler
sees them as the same object
        System.out.println(s1 == s3);           //false because they are
different objects due to the "new String"
        System.out.println(s1.equals(s3));      //true because the value of
the objects is the same
    }
}
```

Output: true false true

Task 4:

```
import java.util.Scanner;

public class Task4{
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        System.out.println("Enter an integer: ");
        String input_String = scan.nextLine();

        try {
            int number = Integer.parseInt(input_String);
            System.out.println(input_String + " is an " + ((number % 2 == 0) ? "even number!" : "odd number!"));
        } catch (NumberFormatException e) {
            System.out.println(input_String + " is not a number.");
        }
    }
}
```

Output: Enter an integer: 15
15 is an odd number!

Task 5:

```
@Override // Modification to Task5
public String toString() {
    return "Customer = " + customer +
        "\nBank = " + bank +
        "\nAccount = " + account +
        "\nBalance = " + balance +
        "\nLimit = " + limit;
}

public static void main(String[] args) {
    Task5[] wallet = new Task5[3];
    wallet[0] = new Task5("John Bowman", "California Savings",
        "5391 0375 9387 5309", 5000);
    wallet[1] = new Task5("John Bowman", "California Federal",
        "3485 0399 3395 1954", 3500);
    wallet[2] = new Task5("John Bowman", "California Finance",
        "5391 0375 9387 5309", 2500, 300);
}
```

```

    for (int val = 1; val <= 16; val++) {
        wallet[0].charge(3*val);
        wallet[1].charge(2*val);
        wallet[2].charge(val);
    }

    for (Task5 card : wallet) {
        System.out.println(card); // Replacing printSummary with println
        while (card.getBalance() > 200.0) {
            card.makePayment(200);
            System.out.println("New balance = " + card.getBalance());
        }
    }
}

```

Output:

Customer = John Bowman
 Bank = California Savings
 Account = 5391 0375 9387 5309
 Balance = 408.0
 Limit = 5000
 New balance = 208.0
 New balance = 8.0
 Customer = John Bowman
 Bank = California Federal
 Account = 3485 0399 3395 1954
 Balance = 272.0
 Limit = 3500
 New balance = 72.0
 Customer = John Bowman
 Bank = California Finance
 Account = 5391 0375 9387 5309
 Balance = 436.0
 Limit = 2500
 New balance = 236.0
 New balance = 36.0

Task 6:

```
// Task6

class Pair<K, V> { // Pair Class

    // Variables
    private K key;
    private V value;

    public Pair(K key, V value) {
        this.key = key;
        this.value = value;
    }

    public K getKey() {
        return key;
    }

    public V getValue() {
        return value;
    }

    public void setKey(K key) {
        this.key = key;
    }

    public void setValue(V value) {
        this.value = value;
    }

    @Override
    public String toString() {
        return "Key: " + key + ", Value: " + value;
    }
}

// Main Class

public class Task6 {
```

```

public static void main(String[] args) {
    Pair<String, Integer> pair1 = new Pair<>("Age", 25);
    Pair<String, Double> pair2 = new Pair<>("Price", 19.99);
    Pair<Integer, String> pair3 = new Pair<>(1, "One");

    System.out.println(pair1);
    System.out.println(pair2);
    System.out.println(pair3);

    pair1.setKey("Updated Age");
    pair1.setValue(30);

    System.out.println(pair1);
}
}

```

Output:

Key: Age, Value: 25
 Key: Price, Value: 19.99
 Key: 1, Value: One
 Key: Updated Age, Value: 30

Task 7:

Output:

C's is 3
 B's is 2
 A's is 4

Explanation:

When the object C c = new C(); is created, each class has its own instance variable x.

A.x = 1
 B.x = 2
 C.x = 3

The statement:

c.superDuperSetIt(4);

calls the method:

super.superSetIt(4);

Inside class C, super refers to class B. Therefore, B's method superSetIt(4) is executed:

```
public void superSetIt(int y) { super.x = y; }
```

In class B, super.x refers to A's variable x. Therefore:

A.x = 4

After execution, the values become:

A.x = 4

B.x = 2

C.x = 3

Now the print statements execute:

c.getIt()
returns C.x = 3

c.superGetIt()
returns B.x = 2

c.superDuperGetIt()
returns A.x = 4

Therefore, the final output is:

C's is 3

B's is 2

A's is 4